Practice workbook · What a log means: reading values and switching to index form

Using the definition of a logarithm to read off a value or to turn one log statement straight into a power.

5 question parts: 1 worked-example retry and 4 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

OLevels.com SCHOOL

Type ID: FPM-LOG-01

1. Question 8 · part b.i · 1 mark

2021 JUN · Paper 01

Worked-example retry · Attempt part b.i only.

8 Given that n satisfies the equation

$$\log_a n = \log_a 3 + \log_a(2n - 1)$$

(a) find the value of n.

(3)

Given that $\log_p x = 3$ and $\log_p y - 3\log_p 2 = 4$

(b) (i) express x in terms of p,

(1)

(ii) express xy in terms of p.

(4)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

1. Question 8 · part b.i · 1 mark · Working and answer

Working and answer

OLevels.com SCHOOL

2. Question 3 · part a · 1 mark

2019 JUN · Paper 01 (regional R)

Further practice · Attempt part a only.

3 (a) Write down the value of $\log_3 9$

(1)

(b) Solve the equation $\log_3 9t = \log_9\left(\frac{12}{t}\right)^2 + 2$ where $t > 0$

Give your answer in the form $a\sqrt{b}$ where a and b are prime numbers.

(6)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com — SCHOOL —

P 6 1 8 8 2 A 0 6 3 6

2. Question 3 · part a · 1 mark · Working and answer

Working and answer

OLevels.com SCHOOL

3. Question 7 · part a · 1 mark

2022 JAN · Paper 01

Further practice · Attempt part a only.

7 (a) Write down the value of $\log_2 16$

(1)

Given that $4 + 2\log_4 x = \log_2 y$

(b) show that $y = 16x$

(4)

(c) Hence solve the equation $4 + 2\log_4 x = \log_2(4x + 5)$

(3)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

3. Question 7 · part a · 1 mark · Working and answer

Working and answer

OLevels.com SCHOOL

4. Question 10 · part i · 2 marks

2025 JUN · Paper 01

Further practice · Attempt part i only.

10 Solve the equation

(i) $\log_4(6y-5)=3$

(2)

(ii) $\log_4(4-3x)^2-\log_2(x^2-5)-3=0$
Show clear algebraic working.
Give your answer to 3 significant figures.

(9)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

32

P 7 2 8 6 5 A 0 3 2 3 6

4. Question 10 · part i · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

5. Question 9 · part a · 2 marks

2024 JUN · Paper 01

Further practice · Attempt part a only.

9 (a) Find the value of a such that $\log_a 8 = \frac{3}{4}$

(2)

(b) Show that

$$3x\log_2 x - 4\log_{16} 8 + 6x\log_4 8 - \log_2 x = \log_2 (8x)^{3x-1}$$

(4)

(c) Hence solve the equation $3x\log_2 x - 4\log_{16} 8 + 6x\log_4 8 - \log_2 x = 0$

(3)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

5. Question 9 · part a · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL